



# Scientific Notation

**Course:** Intermediate Algebra · **Unit 1:** Expressions · **Deck:** U1Lo5\_Scientific\_Notation.pptx (22 slides)

##  You do not need to know algebra to teach this lesson. Read what is written. Follow the slides. Check answers against the key. The learning expectations for today are the same as any other day — this plan gives you the tools to keep the lesson moving. The art of teaching stays with your teacher; the tools are right here.

## QUICK START — BEFORE THE BELL (2 MINUTES)

- ☐ Open the slide deck: **U1Lo5\_Scientific\_Notation.pptx**
- ☐ Write the Bell Ringer on the board (see Slide 1 / Answer Key below)
- ☐ Test the projector
- ☐ Write your name on the board
- ☐ Find the answer key — it is printed inside this plan

## MATERIALS FOR THIS LESSON

- ☐ Whiteboards, markers, erasers (1 set per student)
- ☐ Direct Instruction slide deck (Navy 1C2950 headings, Arial body)
- ☐ Tiered Independent Worksheet (T1/T2/T3 versions)
- ☐ Merit Time: Power of 10 Lane card sets (laminated, zero-device)
- ☐ Exit Ticket slips / LMS exit ticket
- ☐ Visual vocabulary anchor chart
- ☐ Theme visual: Itasca State Park

If anything is missing, check the teacher's desk or ask the front office.

## PACING AT A GLANCE

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| Slide | By minute | Phase                                       |
|-------|-----------|---------------------------------------------|
| 1     | 0–4       | Bell Ringer                                 |
| 7     | 15–16     | End of Direct Instruction                   |
| 19    | 42        | End of Practice (worksheet + game/activity) |
| 22    | 50        | Exit Ticket & Wrap-Up                       |

**Pacing tip:** bell ringer ~4 min · title/I-can/vocab ~4 min · collab ~4 min · worked example + guided practice ~7 min · culture + career ~6 min · GO/SET/EXTRA ~12 min · game/activity ~4 min · individual solve ~4 min · summary + exit ~4 min. Keep the timer visible; the deck drives the pace.

## SUBSTITUTE SCRIPT (SLIDE BY SLIDE)

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### Slide 1 — Bell Ringer

**On screen:** WHITEBOARDS READY! — Tiered prompts (T1 one-step, T2 two-step, T3 challenge) shown on the slide.

**Say:** “Whiteboards ready — three tiers, four minutes. Choose your tier, work silently, show your steps, and hold your board up when the timer hits zero.”

**Do:** Walk the room as they work; cue “3... 2... 1 — boards up” and glance for accuracy.

**Answer key:** Answers are revealed on the slide after the timer. Bridge to today’s topic: Scientific Notation.

### Slide 2 — Title

**On screen:** INTERMEDIATE ALGEBRA — UNIT 1: LESSON 5 — SCIENTIFIC NOTATION. Minnesota standards: 9.3.5.1. MERIT EDUCATION GROUP.

**Say:** “Welcome. Today’s lesson is Unit 1, Lesson 5: Scientific Notation. Everything you need is on these slides — follow along and write when I ask.”

**Do:** Read the lesson title aloud. Pause for students to find a pencil.

### Slide 3 — I Can Statements

**On screen:** I CAN STATEMENTS — Read and write numbers in scientific notation; Convert between standard form and scientific notation; Multiplying and dividing powers of ten.

**Say:** “Here’s what we’ll be able to do by the end of the lesson. Read them silently; we’ll come back to these at the end.”

**Do:** Give 15 seconds, then continue.

### Slide 4 — Vocabulary

**On screen:** VOCABULARY — Scientific notation — a compact way to write very large or very small numbers as a number times a power of ten; Standard form — the usual way of writing a number with all its digits; Power of ten — ten raised to an exponent, such as  $10^3$  or  $10^{-4}$ .

**Say:** “Quick vocabulary check. These are the words for today — I’ll read them, you repeat. [Read each term aloud.] We’ll use them all lesson.”

**Do:** Point to each term as you read it.

### Slide 5 — Collaboration: Think-Write-Pair-Share

**On screen:** ENGAGEMENT: THINK-WRITE-PAIR-SHARE — prompt on the slide, tied to Scientific Notation.

**Say:** “Think-Write-Pair-Share. [Read the prompt aloud.] THINK silently for 30 seconds, then work with your partner or group, then we’ll share out.”

**Do:** Time 30s think, 45s pair, then take 2–3 volunteers.

### Slide 6 — Worked Example (I Do)

**On screen:** WORKED EXAMPLE (I DO) — Step-by-step: Write  $3.2 \times 10^4$  in standard form  $\rightarrow 32,000$ .

**Say:** “A worked example — watch how it’s done step by step. The slide shows every step; I’ll read them as they appear.”

**Do:** Read each step aloud as it is revealed. Answer questions by pointing back to the slide.

### Slide 7 — Guided Practice (We Do)

**On screen:** GUIDED PRACTICE (WE DO) — Together: Write 0.00052 using scientific notation. Multiply  $2 \times 10^3$  by  $3 \times 10^4 \rightarrow 5.2 \times 10^{-4}$ ;  $6 \times 10^7$ .

**Say:** “We do this one together. I’ll ask, you answer — then we check the slide.”

**Do:** Work step by step; call on students for the next step before revealing it.

### Slide 8 — Culture

**On screen:** CULTURE: Ethiopian (Ethiopia). Ethiopian Americans are a growing Minnesota community, centered in Cedar-Riverside. Ethiopian academic tradition dates to antiquity, including the Ge’ez numeral system used for counting and record-keeping.

**Say:** “A cultural connection. [Read the caption aloud.] Notice how math shows up in everyday life and community traditions.”

**Do:** If the slide shows an image only, describe what students see; no math required.

### Slide 9 — Culture Word Problem

**On screen:** CULTURE WORD PROBLEM — An Ethiopian scribe records the number of star-grain seeds counted in Ge’ez as  $4 \times 10^8$ . Write  $4 \times 10^8$  in standard form, and state how many times larger it is than  $4 \times 10^3$ .

**Say:** “Here’s the cultural word problem. Read it to yourself — we’ll solve it together. Use the reading strategy on the slide: circle, underline, box, write.”

**Do:** Give 30 seconds to read; walk through the strategy, then the solution.

### Slide 10 — Career

**On screen:** CAREER: RETAIL OPERATIONS COORDINATOR — Mall of America, Bloomington, MN. Entry wage: \$41,226/year (MN DEED OES, 25th percentile). Median wage: \$50,128/year.

**Say:** “A real career uses this math every day. [Read the career spotlight aloud.] The math we’re learning today is the math people use at work.”

**Do:** Read confidently; the salary and role are on the slide.

### Slide 11 — Career Word Problem

**On screen:** CAREER WORD PROBLEM — An inventory specialist at Best Buy logs  $n$  shipments of televisions, each holding  $1.25 \times 10^3$  units. Write the total number of units in scientific notation and multiply when  $n = 4$ .

**Say:** “Here’s the career problem. Read it to yourself — we’ll solve it together. Use the reading strategy on the slide.”

**Do:** Give 30 seconds; walk the strategy; then the solution.

**Answer key:** Answer is revealed on the slide (solution mirrors today’s practice: 32,000).

### Slide 12 — GO Practice (Your Turn)

**On screen:** GO PRACTICE: YOUR TURN — Write  $3.2 \times 10^4$  in standard form

**Say:** “Your turn — a problem just like the one we did. Work it on your whiteboard (or paper). Two minutes.”

**Do:** Circulate. Cue boards up at 2 minutes.

**Answer key:** 32,000

### Slide 13 — GO Answer

**On screen:** GO ANSWER — reveal: 32,000.

**Say:** “Let’s check it. The slide walks through the solution step by step — I’ll read each step. Check your work against it; fix anything you missed.”

**Do:** Read the solution steps aloud.

### Slide 14 — SET Practice

**On screen:** SET PRACTICE — two problems: Write 0.00052 using scientific notation. Multiply  $2 \times 10^3$  by  $3 \times 10^4$ .

**Say:** “Two practice problems to deepen your understanding. Work them on paper or whiteboard. You have 5 minutes; I’ll be walking around.”

**Do:** Circulate and support. Cue time at 3 and 1 minutes.

### Slide 15 — SET Answer

**On screen:** SET ANSWER — reveal:  $5.2 \times 10^{-4}$ ;  $6 \times 10^7$ .

**Say:** “Let’s check. Read the steps on the slide; self-correct with a pen.”

**Do:** Read the steps aloud; answer questions by re-reading the relevant step.

### Slide 16 — EXTRA Practice Challenge

**On screen:** EXTRA PRACTICE CHALLENGE — A cell phone has  $6.4 \times 10^{10}$  bytes of storage. Write it in standard form

**Say:** “Challenge problem — for anyone who wants a stretch. Try it on your own paper. We’ll go over it in a moment.”

**Do:** Encourage all to attempt; it is optional.

### Slide 17 — EXTRA Answer

**On screen:** EXTRA ANSWER — reveal: 64,000,000,000.

**Say:** “Let’s check the challenge. [Read the steps.] Great work.”

**Do:** Read the solution steps aloud; celebrate attempts.

### Slide 18 — Merit Time Game: Power of 10 Lane

**On screen:** MERIT TIME GAME: POWER OF 10 LANE — SETUP: 4–6 teams, each team gets 16 cards (8 prompts + 8 matches). PLAY: teams race to complete the match. First team with all 8 correct wins! SCORING: 1 point per correct match + 2 bonus points for the first team finished. ZERO-DEVICE · WHOLE-CLASS.

**Say:** “Quick whole-class game — we do this as a team, quietly. I’ll read the rules from the slide.”

**Do:** Read the rules; run the game as the slide directs; keep it positive.

**Game from lesson plan:** Merit Time: Power of 10 Lane (zero-device). Students work in teams to match prompts to answers. Teacher facilitates verification and corrects common errors aloud.

### Slide 19 — Individual Solve: Tiered Worksheet

**On screen:** INDIVIDUAL SOLVE: TIERED WORKSHEET — TIER 1 Build Confidence (6 questions), TIER 2 Apply Your Skills (6 questions), TIER 3 Challenge Yourself (6 questions). PROBLEMS: TIERED WORKSHEET — SCIENTIFIC NOTATION + IN YOUR LMS. SHOW YOUR WORK.

**Say:** “Independent work now — the tiered worksheet. Pick your tier, show your work, and submit in the LMS when you finish.”

**Do:** Circulate; support tier choices; collect or confirm LMS submission.

### Slide 20 — Summary

**On screen:** SUMMARY — RECAP: key ideas from Scientific Notation. REVIEW: the skills practiced today. [Slide lists the takeaways.]

**Say:** “Three big takeaways from today. [Read them aloud.] This is what you should remember tomorrow.”

**Do:** Read each takeaway; pause between them.

## IF A STUDENT ASKS...

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**“Why are we doing this?”** — “This is today’s learning expectation for Scientific Notation. It builds toward what’s next in the unit, and your teacher will pick up right here tomorrow.”

*Redirects (from the lesson plan):*

| Misconception                                          | Redirect Question                                                 | Corrective Strategy                                                               |
|--------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Writes scientific notation with the wrong power of ten | “How many places did the decimal move, and in which direction?”   | Count the place-value shift aloud; match the sign to the direction.               |
| Puts the 1-digit rule after the decimal                | “What digit must come before the decimal in scientific notation?” | Anchor to the format: one non-zero digit, then a decimal point.                   |
| Adds/multiplies powers of ten as plain digits          | “When multiplying powers of ten, what happens to the exponents?”  | Apply the exponent rules: add exponents when multiplying, subtract when dividing. |

## IF A STUDENT FINISHES EARLY

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- Students who finish early do the **Challenge** side of the worksheet — do not invent extra work.
- If the whole class is ahead: re-read the Summary slide and ask students to explain one takeaway in their own words.
- **Do not** move ahead in the curriculum.

## IF THE PROJECTOR DIES

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Write the Bell Ringer problems on the board (from the Answer Key).

Read the worksheet directions aloud; students work in pairs.

Use the printed answer key to check as a class.

The lesson still runs — the script lives in this plan, not the projector.

## WITH A PARAPROFESSIONAL OR CO-TEACHER

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If an adult is in the room supporting students, they know the students best — **follow their lead** on individual support. Check the roster for accommodations; extended time on the Exit Ticket is fine if indicated.

## SUB REPORT (LEAVE FOR THE TEACHER)

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What did we complete? (bell ringer / DI / practice / game / exit ticket)

Who struggled or needed extra support?

Behavior notes (positive first, then concerns)

Anything tech-related that went wrong

## TEACHER'S OWN NOTES

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*(The regular teacher may add specifics here — their timer, signals, roster quirks.)*

*Generated by the MEG Substitute Plan pipeline from the Lesson Plan + Teacher Edition — synced to the published slide decks (2026-08-19).*